

Input Table

Input Table															
	Line		Trafo					Common					Cable		
Core Nr.	Ikmax	Itmax	Irt	Itf	TMVA	Un	Zt	Ipr	ALF	Rct	Sr	Sn	cabel length	cable diameter	max Temperatur
1	50,0	50,0	0,5	2,0	220	220	23,0	800	20	7	0,02	30	90	4	75
1	6,6	50,0	1,2	6,6	310	150	18,0	1250	20	7	0,02	10	65	4	75
2	29,0	40,0	1,3	29,0	325	150	5,0	1250	20	7	0,02	10	65	4	75
3	40,0	40,0	1,2	8,0	310	150	18,0	1250	20	7,5	0,02	30	75	4	75
4	40,0	40,0	0,8	6,0	200	150	15,0	1600	40	7,5	0,02	10	70	4	75
1	31,5	31,5	2,0	16,0	110	33	12,0	2500	20	14	0,02	15	13	1,5	75
2	40,0	40,0	2,7	29,0	175	33	12,0	3000	60	20	0,02	30	110	4	75
3	29,0	30,0	2,7	25,0	155	33	12,0	2500	20	14	0,02	20	55	4	75
4	29,0	29,0	2,7	25,0	155	33	10,0	3000	20	18	0,02	20	80	4	75
5	8,4	8,4	0,5	6,0	30	33	8,0	500	20	2	0,02	10	40	2,5	75

220 kV - Core 1		
General parameters		
Cable length	90,00	m
Cable cross-section	4,00	mm ²
Maximum temperature	75,00	°C
CT secondary resistance R _{ct}	7,00	Ohm
CT rated primary current I _{pr}	800,00	A
CT rated secondary current I _{sr}	1,00	A
IED input burden S _R	0,02	VA
Accuracy limiting factor ALF	20,00	
Rated CT output S _n	30,00	VA
Intermediate calculations		
Total secondary burden	7,9777	Ohm
Available CT E _{al}	740,0000	V
RET670 - CT Requirement for Transformer Differential Protection		
Inputs		
Transformer Rating S _t	220,00	MVA
Nominal system voltage	220,00	kV
Transformer impedance	23,00	%
Intermediate calculations		
Transformer rated primary current I _{rt}	577,3503	A
Max. current through two main CTs and transformer I _{tf}	2.510,2186	A
Results		
E _{alreq} criterion 1: transformer rated current	172,7227	V
E _{alreq} criterion 2: through-fault current	50,0645	V
Required E _{al}	172,7227	V
Pass / fail	pass	
RED670 - CT Requirement for Line Differential Protection		
Inputs		
Maximum internal close-in fault current I _{kmax}	50.000	A
Maximum external through-fault current I _{tmax}	50.000	A
Results		
E _{alreq} equation 601: internal close-in fault	498,6074	V
E _{alreq} equation 602: external through-fault	997,2148	V
Required E _{al}	997,2148	V
Pass / fail	fail	

150 kV - Core 1		
General parameters		
Cable length	65,00	m
Cable cross-section	4,00	mm ²
Maximum temperature	75,00	°C
CT secondary resistance R _{ct}	7,00	Ohm
CT rated primary current I _{pr}	1.250,00	A
CT rated secondary current I _{sr}	1,00	A
IED input burden S _R	0,02	VA
Accuracy limiting factor ALF	20,00	
Rated CT output S _n	10,00	VA
Intermediate calculations		
Total secondary burden	7,7117	Ohm
Available CT E _{al}	340,0000	V
RET670 - CT Requirement for Transformer Differential Protection		
Inputs		
Transformer Rating S _t	310,00	MVA
Nominal system voltage	150,00	kV
Transformer impedance	18,00	%
Intermediate calculations		
Transformer rated primary current I _{rt}	1.193,1906	A
Max. current through two main CTs and transformer I _{tf}	6.628,8364	A
Results		
E _{alreq} criterion 1: transformer rated current	220,8362	V
E _{alreq} criterion 2: through-fault current	81,7912	V
Required E _{al}	220,8362	V
Pass / fail	pass	
RED670 - CT Requirement for Line Differential Protection		
Inputs		
Maximum internal close-in fault current I _{kmax}	6.600	A
Maximum external through-fault current I _{tmax}	50.000	A
Results		
E _{alreq} equation 601: internal close-in fault	40,7177	V
E _{alreq} equation 602: external through-fault	616,9348	V
Required E _{al}	616,9348	V
Pass / fail	fail	

150 kV - Core 2

General parameters

Cable length	65,00	m
Cable cross-section	4,00	mm ²
Maximum temperature	75,00	°C
CT secondary resistance R _{ct}	7,00	Ohm
CT rated primary current I _{pr}	1.250,00	A
CT rated secondary current I _{sr}	1,00	A
IED input burden S _R	0,02	VA
Accuracy limiting factor ALF	20,00	
Rated CT output S _n	10,00	VA

Intermediate calculations

Total secondary burden	7,7117	Ohm
Available CT E _{al}	340,0000	V

RET670 - CT Requirement for Transformer Differential Protection

Inputs

Transformer Rating S _t	325,00	MVA
Nominal system voltage	150,00	kV
Transformer impedance	5,00	%

Intermediate calculations

Transformer rated primary current I _{rt}	1.250,9256	A
Max. current through two main CTs and transformer I _{tf}	25.018,5117	A

Results

E _{alreq} criterion 1: transformer rated current	231,5219	V
E _{alreq} criterion 2: through-fault current	308,6958	V
Required E _{al}	308,6958	V
Pass / fail	pass	

RED670 - CT Requirement for Line Differential Protection

Inputs

Maximum internal close-in fault current I _{kmax}	29.000	A
Maximum external through-fault current I _{tmax}	40.000	A

Results

E _{alreq} equation 601: internal close-in fault	178,9111	V
E _{alreq} equation 602: external through-fault	493,5479	V
Required E _{al}	493,5479	V
Pass / fail	fail	

150 kV - Core 3		
General parameters		
Cable length	75,00	m
Cable cross-section	4,00	mm ²
Maximum temperature	75,00	°C
CT secondary resistance R _{ct}	7,50	Ohm
CT rated primary current I _{pr}	1.250,00	A
CT rated secondary current I _{sr}	1,00	A
IED input burden S _R	0,02	VA
Accuracy limiting factor ALF	20,00	
Rated CT output S _n	30,00	VA
Intermediate calculations		
Total secondary burden	8,3181	Ohm
Available CT E _{al}	750,0000	V
RET670 - CT Requirement for Transformer Differential Protection		
Inputs		
Transformer Rating S _t	310,00	MVA
Nominal system voltage	150,00	kV
Transformer impedance	18,00	%
Intermediate calculations		
Transformer rated primary current I _{rt}	1.193,1906	A
Max. current through two main CTs and transformer I _{tf}	6.628,8364	A
Results		
E _{alreq} criterion 1: transformer rated current	238,2018	V
E _{alreq} criterion 2: through-fault current	88,2229	V
Required E _{al}	238,2018	V
Pass / fail	pass	
RED670 - CT Requirement for Line Differential Protection		
Inputs		
Maximum internal close-in fault current I _{kmax}	40.000	A
Maximum external through-fault current I _{tmax}	40.000	A
Results		
E _{alreq} equation 601: internal close-in fault	266,1792	V
E _{alreq} equation 602: external through-fault	532,3583	V
Required E _{al}	532,3583	V
Pass / fail	pass	

150 kV - Core 4		
General parameters		
Cable length	70,00	m
Cable cross-section	4,00	mm ²
Maximum temperature	75,00	°C
CT secondary resistance R _{ct}	7,50	Ohm
CT rated primary current I _{pr}	1.600,00	A
CT rated secondary current I _{sr}	1,00	A
IED input burden S _R	0,02	VA
Accuracy limiting factor ALF	40,00	
Rated CT output S _n	10,00	VA
Intermediate calculations		
Total secondary burden	8,2649	Ohm
Available CT E _{al}	700,0000	V
RET670 - CT Requirement for Transformer Differential Protection		
Inputs		
Transformer Rating S _t	200,00	MVA
Nominal system voltage	150,00	kV
Transformer impedance	15,00	%
Intermediate calculations		
Transformer rated primary current I _{rt}	769,8004	A
Max. current through two main CTs and transformer I _{tf}	5.132,0024	A
Results		
E _{alreq} criterion 1: transformer rated current	119,2934	V
E _{alreq} criterion 2: through-fault current	53,0193	V
Required E _{al}	119,2934	V
Pass / fail	pass	
RED670 - CT Requirement for Line Differential Protection		
Inputs		
Maximum internal close-in fault current I _{kmax}	40.000	A
Maximum external through-fault current I _{tmax}	40.000	A
Results		
E _{alreq} equation 601: internal close-in fault	206,6223	V
E _{alreq} equation 602: external through-fault	413,2446	V
Required E _{al}	413,2446	V
Pass / fail	pass	

33 kV - Core 1

General parameters

Cable length	13,00	m
Cable cross-section	1,50	mm ²
Maximum temperature	75,00	°C
CT secondary resistance R _{ct}	14,00	Ohm
CT rated primary current I _{pr}	2.500,00	A
CT rated secondary current I _{sr}	1,00	A
IED input burden S _R	0,02	VA
Accuracy limiting factor ALF	20,00	
Rated CT output S _n	15,00	VA

Intermediate calculations

Total secondary burden	14,3889	Ohm
Available CT E _{al}	580,0000	V

RET670 - CT Requirement for Transformer Differential Protection

Inputs

Transformer Rating S _t	110,00	MVA
Nominal system voltage	33,00	kV
Transformer impedance	12,00	%

Intermediate calculations

Transformer rated primary current I _{rt}	1.924,5009	A
Max. current through two main CTs and transformer I _{tf}	16.037,5075	A

Results

E _{alreq} criterion 1: transformer rated current	332,2974	V
E _{alreq} criterion 2: through-fault current	184,6097	V
Required E _{al}	332,2974	V
Pass / fail	pass	

RED670 - CT Requirement for Line Differential Protection

Inputs

Maximum internal close-in fault current I _{kmax}	31.500	A
Maximum external through-fault current I _{tmax}	31.500	A

Results

E _{alreq} equation 601: internal close-in fault	181,3001	V
E _{alreq} equation 602: external through-fault	362,6003	V
Required E _{al}	362,6003	V
Pass / fail	pass	

33 kV - Core 2

General parameters

Cable length	110,00	m
Cable cross-section	4,00	mm ²
Maximum temperature	75,00	°C
CT secondary resistance R _{ct}	20,00	Ohm
CT rated primary current I _{pr}	3.000,00	A
CT rated secondary current I _{sr}	1,00	A
IED input burden S _R	0,02	VA
Accuracy limiting factor ALF	60,00	
Rated CT output S _n	30,00	VA

Intermediate calculations

Total secondary burden	21,1905	Ohm
Available CT E _{al}	3.000,0000	V

RET670 - CT Requirement for Transformer Differential Protection

Inputs

Transformer Rating S _t	175,00	MVA
Nominal system voltage	33,00	kV
Transformer impedance	12,00	%

Intermediate calculations

Transformer rated primary current I _{rt}	3.061,7060	A
Max. current through two main CTs and transformer I _{tf}	25.514,2164	A

Results

E _{alreq} criterion 1: transformer rated current	648,7922	V
E _{alreq} criterion 2: through-fault current	360,4401	V
Required E _{al}	648,7922	V
Pass / fail	pass	

RED670 - CT Requirement for Line Differential Protection

Inputs

Maximum internal close-in fault current I _{kmax}	40.000	A
Maximum external through-fault current I _{tmax}	40.000	A

Results

E _{alreq} equation 601: internal close-in fault	282,5406	V
E _{alreq} equation 602: external through-fault	565,0812	V
Required E _{al}	565,0812	V
Pass / fail	pass	

33 kV - Core 3

General parameters

Cable length	55,00	m
Cable cross-section	4,00	mm ²
Maximum temperature	75,00	°C
CT secondary resistance R _{ct}	14,00	Ohm
CT rated primary current I _{pr}	2.500,00	A
CT rated secondary current I _{sr}	1,00	A
IED input burden S _R	0,02	VA
Accuracy limiting factor ALF	20,00	
Rated CT output S _n	20,00	VA

Intermediate calculations

Total secondary burden	14,6053	Ohm
Available CT E _{al}	680,0000	V

RET670 - CT Requirement for Transformer Differential Protection

Inputs

Transformer Rating S _t	155,00	MVA
Nominal system voltage	33,00	kV
Transformer impedance	12,00	%

Intermediate calculations

Transformer rated primary current I _{rt}	2.711,7967	A
Max. current through two main CTs and transformer I _{tf}	22.598,3060	A

Results

E _{alreq} criterion 1: transformer rated current	475,2784	V
E _{alreq} criterion 2: through-fault current	264,0435	V
Required E _{al}	475,2784	V
Pass / fail	pass	

RED670 - CT Requirement for Line Differential Protection

Inputs

Maximum internal close-in fault current I _{kmax}	29.000	A
Maximum external through-fault current I _{tmax}	30.000	A

Results

E _{alreq} equation 601: internal close-in fault	169,4212	V
E _{alreq} equation 602: external through-fault	350,5265	V
Required E _{al}	350,5265	V
Pass / fail	pass	

33 kV - Core 4

General parameters

Cable length	80,00	m
Cable cross-section	4,00	mm ²
Maximum temperature	75,00	°C
CT secondary resistance R _{ct}	18,00	Ohm
CT rated primary current I _{pr}	3.000,00	A
CT rated secondary current I _{sr}	1,00	A
IED input burden S _R	0,02	VA
Accuracy limiting factor ALF	20,00	
Rated CT output S _n	20,00	VA

Intermediate calculations

Total secondary burden	18,8713	Ohm
Available CT E _{al}	760,0000	V

RET670 - CT Requirement for Transformer Differential Protection

Inputs

Transformer Rating S _t	155,00	MVA
Nominal system voltage	33,00	kV
Transformer impedance	10,00	%

Intermediate calculations

Transformer rated primary current I _{rt}	2.711,7967	A
Max. current through two main CTs and transformer I _{tf}	27.117,9672	A

Results

E _{alreq} criterion 1: transformer rated current	511,7514	V
E _{alreq} criterion 2: through-fault current	341,1676	V
Required E _{al}	511,7514	V
Pass / fail	pass	

RED670 - CT Requirement for Line Differential Protection

Inputs

Maximum internal close-in fault current I _{kmax}	29.000	A
Maximum external through-fault current I _{tmax}	29.000	A

Results

E _{alreq} equation 601: internal close-in fault	182,4226	V
E _{alreq} equation 602: external through-fault	364,8452	V
Required E _{al}	364,8452	V
Pass / fail	pass	

33 kV - Core 5

General parameters

Cable length	40,00	m
Cable cross-section	2,50	mm ²
Maximum temperature	75,00	°C
CT secondary resistance R _{ct}	2,00	Ohm
CT rated primary current I _{pr}	500,00	A
CT rated secondary current I _{sr}	1,00	A
IED input burden S _R	0,02	VA
Accuracy limiting factor ALF	20,00	
Rated CT output S _n	10,00	VA

Intermediate calculations

Total secondary burden	2,7010	Ohm
Available CT E _{al}	240,0000	V

RET670 - CT Requirement for Transformer Differential Protection

Inputs

Transformer Rating S _t	30,00	MVA
Nominal system voltage	33,00	kV
Transformer impedance	8,00	%

Intermediate calculations

Transformer rated primary current I _{rt}	524,8639	A
Max. current through two main CTs and transformer I _{tf}	6.560,7985	A

Results

E _{alreq} criterion 1: transformer rated current	85,0608	V
E _{alreq} criterion 2: through-fault current	70,8840	V
Required E _{al}	85,0608	V
Pass / fail	pass	

RED670 - CT Requirement for Line Differential Protection

Inputs

Maximum internal close-in fault current I _{kmax}	8.400	A
Maximum external through-fault current I _{tmax}	8.400	A

Results

E _{alreq} equation 601: internal close-in fault	45,3775	V
E _{alreq} equation 602: external through-fault	90,7551	V
Required E _{al}	90,7551	V
Pass / fail	pass	

CT Calculation Results & SECP Comparison

Calculation sheet	Protection function	Excel criterion 1 (V)	Excel criterion 2 (V)	Excel proposed Eal (V)	Excel status	SECP criterion 1 (V)	SECP criterion 2 (V)	SECP proposed Eal (V)	SECP status	δ
220 - 1	Line differential	498,61	997,21	740,00	fail	501,83	1.003,65	740,00	fail	0,6 %
150 - 1	Line differential	40,72	616,93	340,00	fail	40,91	619,90	340,00	fail	0,5 %
150 - 2	Line differential	178,91	493,55	340,00	fail	179,77	495,92	340,00	fail	0,5 %
150 - 3	Line differential	266,18	532,36	750,00	pass	267,55	535,10	750,00	pass	0,5 %
150 - 4	Line differential	206,62	413,24	700,00	pass	207,62	415,25	700,00	pass	0,5 %
33 - 1	Line differential	181,30	362,60	580,00	pass	181,47	362,95	580,00	pass	0,1 %
33 - 2	Line differential	282,54	565,08	3.000,00	pass	283,38	566,76	3.000,00	pass	0,3 %
33 - 3	Line differential	169,42	350,53	680,00	pass	169,79	351,28	680,00	pass	0,2 %
33 - 4	Line differential	182,42	364,85	760,00	pass	182,86	365,73	760,00	pass	0,2 %
33 - 5	Line differential	45,38	90,76	240,00	pass	46,05	92,09	240,00	pass	1,5 %
220 - 1	Transformer differential	172,72	50,06	740,00	pass	195,69	56,72	740,00	pass	13,3 %
150 - 1	Transformer differential	220,84	81,79	340,00	pass	242,77	89,91	340,00	pass	9,9 %
150 - 2	Transformer differential	231,52	308,70	340,00	pass	254,52	339,36	340,00	pass	9,9 %
150 - 3	Transformer differential	238,20	88,22	750,00	pass	263,51	97,60	750,00	pass	10,6 %
150 - 4	Transformer differential	119,29	53,02	700,00	pass	131,20	58,31	700,00	pass	10,0 %
33 - 1	Transformer differential	332,30	184,61	580,00	pass	341,45	189,69	580,00	pass	2,8 %
33 - 2	Transformer differential	648,79	360,44	3.000,00	pass	688,48	382,49	3.000,00	pass	6,1 %
33 - 3	Transformer differential	475,28	264,04	680,00	pass	496,37	275,76	680,00	pass	4,4 %
33 - 4	Transformer differential	511,75	341,17	760,00	pass	537,32	358,21	760,00	pass	5,0 %
33 - 5	Transformer differential	85,06	70,88	240,00	pass	109,02	90,85	240,00	pass	28,2 %